Experiment 9: Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objectives, MATLAB Programs and Output Images

Objective 1

Analyze Channel Capacity (bps/Hz) Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

MATLAB Program

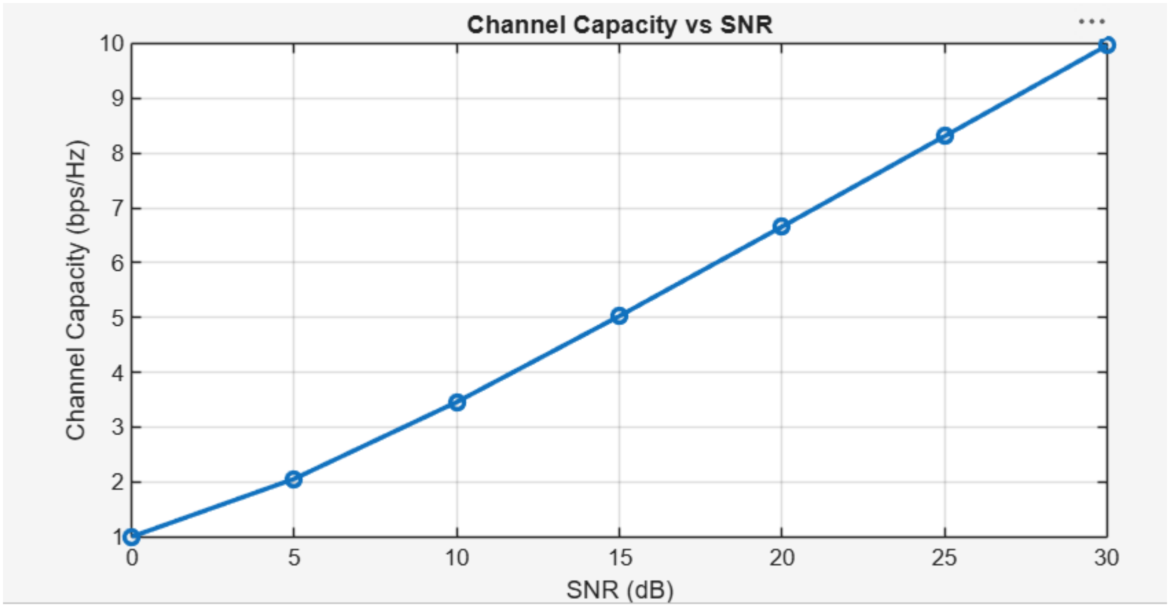
```
clc;
clear;
close all;

snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output Image – Objective 1



Objective 2

Compare Signal-to-Noise Ratio (SNR) (dB) Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

MATLAB Program

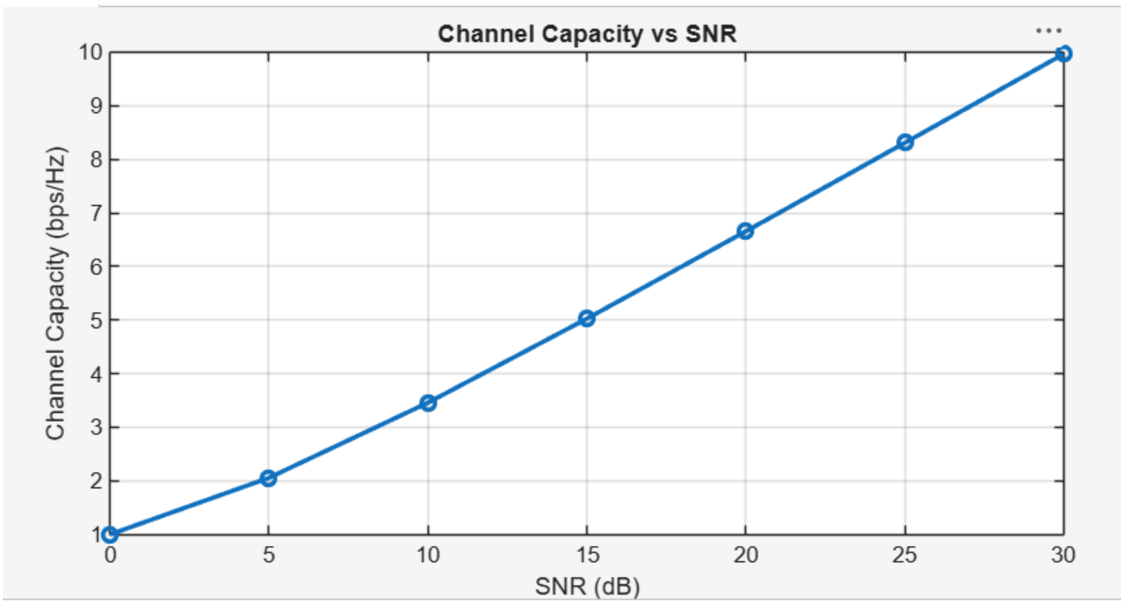
```
clc;
clear;
close all;

snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output Image – Objective 2



Objective 3

Analyze Antenna Gain (dB) Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

MATLAB Program

```
clc;
clear;
close all;

antennas = [4 8 16 32 64];          % Number of Antenna Elements
gain = 10*log10(antennas);          % Antenna Gain (dB)

figure
bar(antennas, gain)

grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```

Output Image – Objective 3